

SHREYA NARAYAN

shreyanarayan@gmail.com | [LinkedIn Profile](#) | [Website](#)

EDUCATION

M.S., Stanford University | Electrical Engineering

Conferred June 2023

Graduate Student Researcher in the Dahl Lab, Arbabian Lab

2021 - 2023

Focus: Ultrasound Imaging, Acoustic Wave Propagation, Signal Processing, Applied ML

- Designed **differentiable beamforming based optimization** using phase error and image entropy minimization, and coherence factor and speckle brightness maximization, for **flexible ultrasound** array geometry estimation in wearable imaging to **maximize resolution** with patient movement.
 - < 0.6 mm RMSE achieved with phase error minimization.
 - Flexible arrays designed and tested in MATLAB Field II. Highlighted work at IUS 2023.
- Designed **signal processing algorithms** for a perceptual audio codec using QMF filter bank and psychoacoustic model, optimized for human auditory perception and compression efficiency.
- Developed **adaptive radar waveform optimization** and **EKF-based multi-target tracking algorithms** for real-time pedestrian gait classification, achieving >90% tracking accuracy in multi-pedestrian environments.
- **Machine Learning on Image Data**
 - Evaluated U-Net + VGG-16 for segmentation of muscle fibers in ultrasound images, 0.99 IOU
 - Reduced MRNet model training time by 84% using reduced feature space (33% of the training data) with insignificant effect on accuracy for identification of ACL tears in MRI Imaging

B.S., Johns Hopkins University | Biomedical Engineering

Conferred May 2021

Minors in Entrepreneurship & Management and Music

[Clark Scholar](#)

EXPERIENCE

Greeneridge Sciences, Inc. | Acoustics Systems Engineer

September 2024 - Present

- Designed, developed, and deployed an [end-to-end system and pipeline](#) for detection, classification, and localization of whale calls from directional autonomous seafloor acoustic recorders (DASARs), by doing the following:
 - **Developed signal processing pipelines** to convert raw acoustic data into time-frequency representations and structured inputs ideal for automated analysis.
 - **Designed and implemented ML models** including a ResNet50 Multilabel classifier for identification of whale calls (97.3% accuracy) and Faster-RCNN detection system (95% precision, 84% recall) for detecting (boxing) calls and classifying call type.
 - **Designed and validated a localization algorithm (OpenCV)** that achieved a median error of 250 m across > 400km², well within production error margin. Developed a **Hungarian Matching Algorithm** for robust evaluation and comparison with ground truth.
 - Conducted field work for sensor calibration, testing, deployment, retrieval in the Arctic Ocean.
 - Owned **system integration, testing, and debugging** of the full pipeline, identifying and addressing failure modes. My system scaled to 15M images from 19 DASARs deployed continuously for >2 months, replacing infeasible manual analysis and saving an estimated >\$500K in 2025 and >\$50K annually.
- Developed an interactive GUI for visualization and analysis of sound source directionality, including long term spectral and bearing averaging, enabling efficient data contextualization and understanding of acoustic sources during data analysis.

Shreya Narayan

shreyanarayan@gmail.com | [LinkedIn Profile](#) | [Website](#)

Hubbs Sea World Research Institute | Acoustics Research Engineer

August 2024 - Present

- Designed and evaluated **machine learning and signal processing pipelines** to [analyze acoustic features](#) in vocalizations across species by designing classifiers and autoencoders.
- Applied dimensionality reduction (t-SNE, UMAP), data science, and deep learning interpretability techniques (CAM) for analysis of structured acoustic signals.

CraniUS | Engineering System Design Lead

Jan 2020 - Jan 2022

- Led product development of an implantable ultrasound device for early brain tumor detection, from user needs identification (user stories) to **system architecture development, transducer design, and prototyping**.
- **Translated user requirements into engineering specifications** across hardware and software.
- **Directed a 7-person cross-functional team** and coordinated with neurosurgeons at the Johns Hopkins Hospital to validate design requirements and clinical relevance.
- Secured \$78K in funding and guided iterative design dev. through testing & stakeholder feedback.

University of California, San Diego | Wearable Ultrasound Startup: Product Lead

Jun 2023 - Mar 2024

- Led **user needs discovery** through 15+ interviews with physicians, patients, and stakeholders; translated clinical and workflow requirements into product and physiological sensing specifications
- Developed regulatory, reimbursement, and product strategy through cross-functional collaboration with clinicians and engineers; wrote the team's \$2M NIH STTR proposal.

SKILLS

Technical Expertise: Acoustic signal processing, digital signal processing, systems engineering, machine learning and deep learning (DNN, CNNs), computer vision, acoustics, bioacoustics, time-of-flight sensing and imaging (including underwater/ocean acoustics, radar, medical ultrasound, beamforming), data science, data analysis, and data visualization (tSNE, UMAP, ML Interpretability - Class Activation Mapping), medical imaging, medical device development, oceanographic field work

Technical Languages/Programs: Python (OpenCV, Scipy, Pandas, Matplotlib, Pillow, Numpy, PyTorch), MATLAB, C++, Raven Pro, Solidworks, GitHub, Streamlit

Collaboration & Execution: Technical communication, cross-functional collaboration, strategy development, fundraising, grant writing, project management, team management

AWARDS AND MEMBERSHIPS

Climate Change Learning for Action Fellow (Terra.do), Featured Fellow	2026
Joint Conference: Acoustical Society of America/Japan: Lecture Presentation	2025
Acoustical Society of America, Member	2025 - Present
Highlighted work: Flexible Array Shape Estimation (2023 International Ultrasonics Symposium)	2023
Best ADC Design, Stanford University (EE 315 Design Contest)	2022
Cohen Fund Grant Recipient (Project lead, award of \$78,000), Johns Hopkins	2021
Johns Hopkins Residential Advisor Above and Beyond Award	2020
Johns Hopkins Biomedical Engineering Distinguished Service Award	2019
TCO Labs, Grant Recipient	2019
Clark Scholar Study Abroad Scholarship (Johns Hopkins University, Israel)	2019
Malinow Family Fund Grant	2018
Johns Hopkins Dean's List	2017 - 2021

VOLUNTEER AND LEADERSHIP EXPERIENCE

Acoustical Society of America San Francisco Chapter, Guest Lecturer	2025
California Coast Classic Rider, Arthritis Foundation	2025
Project Lead: Stanford's Beyond the Farm Service Days (Tree planting with Canopy)	2025, 2026
Climate Change Action Oriented Discussion Group Lead	2023 - 2025
Residential Advisor/CA, Stanford University	2022 - 2023
Residential Advisor, Johns Hopkins	2019 - 2021
Running Club Coach, Johns Hopkins	2019 - 2021
Clark Scholar, Johns Hopkins	2017 - 2021
Uganda: Biomedical Engineering Needs Finding and Analysis Program	2020
Israel: Innovation and Entrepreneurship Program	2019

Publications

1. **Narayan, S.**, Bowles A., Ames A. [*Manuscript In Progress*] Unsupervised Machine Learning Techniques for Quantitative Demonstration of Vocal Production Learning in Beluga Whales. Hubbs Sea World Research Institute.
2. **Narayan, S.**, Kim, K., Blackwell, S. [*Manuscript In Progress*] Results from the application of a machine learning based pipeline for detection, classification, and localization of bowhead whale calls from a 19 sensor DASAR array and comparison to manual methods. Greeneridge Sciences.
3. Kim, K.H., Blackwell, S.B., **Narayan S.V** (eds.). 2026. Monitoring of industrial sounds and bowhead whales near Hilcorp's Northstar Oil Development, Alaskan Beaufort Sea, 2024: Summary Report. GSI Rep. 557-1. Rep. from Greeneridge Sciences Inc. (Santa Barbara, CA) for Hilcorp Alaska LLC, Anchorage, AK.
4. **Narayan, S.**, Kim, K. Machine learning based detection, classification, and localization of bowhead whale calls during their annual fall migration across the Beaufort Sea. Sixth Joint Meeting, Acoustical Society of America and Acoustical Society of Japan. *Acoustical Society of America*, 2025 December 1 - 5. Honolulu, Hawaii.
5. D. Hyun, **S. Narayan**, W. Simson, L. Zhuang and J. Dahl. Flexible Array Shape Estimation Using Differentiable Beamforming. 2023 IEEE International Ultrasonics Symposium (IUS), Montreal, Canada, 2023.
6. **Narayan, S.** Evaluation of U-Net vs U-Net + VGG-16 Architecture Performance in Extracting Masks and Musculoskeletal Metrics from Ultrasound B-Mode Image Data [*Unpublished Manuscript*]. 2022. CS230 Deep Learning, Department of Computer Science, Stanford University.
7. Ojiaku CA, Chung E, Parikh V, Williams JK, Schwab A, Fuentes AL, Corpuz ML, Lui V, Paek S, Bexiga NM, **Narayan S**, Nunez FJ, Ahn K, Ostrom RS, An SS, Panettieri RA Jr. Transforming Growth Factor- β 1 Decreases β 2-Agonist-induced Relaxation in Human Airway Smooth Muscle. *Am J Respir Cell Mol Biol*. 2019 Aug;61(2):209-218. doi: 10.1165/rcmb.2018-0301OC. PMID: 30742476; PMCID: PMC6670035.